

Effects of Pacing Strategy on Work Done above Critical Power during High-Intensity Exercise

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ABSTRACT

CHIDNOK, W., F. J. DIMENNA, S. J. BAILEY, D. P. WILKERSON, A. VANHATALO, and A. M. JONES. Effects of Pacing Strategy on Work Done above Critical Power during High-Intensity Exercise. *Med. Sci. Sports Exerc.*, Vol. 45, No. 7, pp. 1377–1385, 2013. **Purpose:** We investigated the influence of pacing strategy on the work completed above critical power (CP) before exhaustion ($W_{>CP}$) and the peak $\dot{V}O_2$ attained during high-intensity cycling. **Methods:** After the determination of $\dot{V}O_{2max}$ from a ramp incremental cycling (INC) test and the estimation of the parameters of the power–duration relationship for high-intensity exercise (i.e., CP and W') from a 3-min all-out cycling test (AOT), eight male subjects completed a cycle test to exhaustion at a severe-intensity constant work rate (CWR) estimated to result in exhaustion in 3 min and a self-paced 3-min cycling time trial (SPT). **Results:** The $\dot{V}O_{2max}$ determined from INC was 4.24 ± 0.69 L·min⁻¹, and the CP and the W' estimated from AOT were 260 ± 60 W and 16.5 ± 4.0 kJ, respectively. $W_{>CP}$ during SPT was not significantly different from $W_{>CP}$ during CWR (15.3 ± 5.6 and 16.6 ± 7.4 kJ, respectively), and these values were also similar to $W_{>CP}$ during INC (16.4 ± 4.0 kJ) and W' estimated from AOT. The peak $\dot{V}O_2$ during SPT was not significantly different from peak $\dot{V}O_2$ during CWR (4.20 ± 0.77 and 4.14 ± 0.75 L·min⁻¹, respectively), and these values were similar to the $\dot{V}O_{2max}$ determined from INC and the peak $\dot{V}O_2$ during AOT (4.10 ± 0.79 L·min⁻¹). **Conclusion:** Exhaustion during high-intensity exercise coincides with the achievement of the same peak $\dot{V}O_2$ ($\dot{V}O_{2max}$) and the completion of the same $W_{>CP}$, irrespective of the work rate forcing function (INC or CWR) or pacing strategy (enforced pace or self-paced). These findings indicate that exhaustion during high-intensity exercise is based on highly predictable physiological processes, which are unaffected when pacing strategy is self-selected. **Key Words:** CRITICAL POWER, W' , SELF-PACED, CONSTANT WORK RATE, FATIGUE

It is well established that the limit of tolerance during high-intensity constant work rate (CWR) exercise conforms to a hyperbolic work rate/time function (26,27) (for a review, see Jones et al. [16]). For example, during cycle exercise at a power output (P) above what has been termed the “critical power” (CP; i.e., the asymptote of the power/time hyperbola), time to exhaustion (T_e) can be predicted as follows:

$$T_e = W' / (P - CP) \quad [1]$$

where W' represents the hyperbola’s curvature constant. This constancy of W' for supra-CP work rates predicts that

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T_e during high-intensity exercise will coincide with the depletion of a fixed capacity for work above CP ($W_{>CP}$) (12,16,26). Recent research indicates that the same model applies to exhaustive exercise where work rate is not held constant, for example, “ramp” incremental cycling (INC) where work rate is increased as a linear function of time (e.g., 1 W every 2 s) (28,30). For this type of exercise, T_e is predicted by a modified version of equation 1:

$$T_e = CP/S + \text{sqrt}(2W'/S) \quad [2]$$

where S represents the ramp slope (e.g., $0.5 \text{ W} \cdot \text{s}^{-1}$) (28). This indicates that regardless of how an exercise bout is configured, the time spent working above CP (e.g., from exercise onset for CWR and once CP is surpassed during ramp INC) is limited by the same finite energetic reserve. The CP model also predicts T_e during intermittent cycling where bouts of high-intensity exercise are interspersed with recovery intervals at reduced work rates (6).

Unlike the aforementioned testing protocols where structured work rate forcing functions are applied by an investigator (i.e., pace is enforced), athletes in sports competitions have the opportunity for self-pacing, and this is known to

have important implications for performance (1,2,9,11). Pacing strategies vary according to event duration and an athlete's level of experience but are believed to reflect an attempt to optimize performance without incurring premature or intolerable challenges to homeostasis (35). Interestingly, there is evidence to suggest that self-pacing reduces the metabolic stress associated with a given performance. For example, it has been shown that allowing subjects to self-pace at a fixed RPE enabled them to complete a 5000-m rowing bout in the same amount of time, but with less physiological perturbation, compared with enforced-pace CWR exercise where the same mean power output was maintained (21). Given that CWR exercise above CP is theoretically terminated when the W' is fully used and $\dot{V}O_{2\max}$ reaches its maximum value (4,16,33), it may be speculated that self-pacing increases the W' and/or the $\dot{V}O_{2\max}$, but this possibility has not been investigated.

In addition to exhaustive high-intensity CWR and INC, all-out sprint exercise results in the attainment of $\dot{V}O_{2\max}$ as long as it is maintained for a sufficient period (13). For example, when subjects cycle all-out against a fixed resistance for 3 min, the peak $\dot{V}O_2$ (typically attained ~60 s into the test and maintained from that point onwards) is not different from the $\dot{V}O_{2\max}$ measured during exhaustive INC or CWR exercise in the severe-intensity domain (3,40). Interestingly, although power output (which is solely a function of cadence within this paradigm) declines precipitously throughout the initial stages of this test, it ultimately reaches a nadir (i.e., end-test power; EP) that is not significantly different from the CP derived from conventional testing procedures (3,37,38). It has also been shown that the total work performed above EP during this 3-min AOT (WEP) approximates W' (37,38). Although it has been reported that the EP and the WEP can be used to accurately predict T_e during CWR (40), it is presently not clear if the same is true for T_e during INC (i.e., via equations 1 and 2). Also, although it has been reported that the CP is significantly correlated with cycling time trial performance (34), the degree to which EP and WEP can together predict performance during self-paced high-intensity exercise is not known.

The primary purpose of this investigation was to determine whether exhaustion during high-intensity exercise coincides with completion of the same $W_{>CP}$ and attainment of the same peak $\dot{V}O_2$ (i.e., a reproducible W' and $\dot{V}O_{2\max}$, respectively) during enforced-pace CWR, self-paced 3-min cycling time trial (SPT) at a similar mean power output, INC, and all-out sprint cycling. We hypothesized that subjects would achieve the same peak $\dot{V}O_2$ and complete the same amount of work in excess of CP at the point at which exercise was terminated irrespective of the test protocol. A secondary purpose was to examine how well parameters derived from a 3-min all-out cycling test (AOT) can predict performance during exhaustive exercise tests using different work rate forcing functions. Specifically, we hypothesized that EP and WEP could be used to accurately predict T_e for CWR and INC and performance during SPT.

METHODS

Subjects. Eight male subjects (mean $\pm$ SD; age = 21 $\pm$ 4 yr, stature = 1.75 $\pm$ 0.11 m, body mass = 76.0 $\pm$ 10.0 kg) volunteered and gave written informed consent to participate in this study, which had been approved by the University of Exeter Research Ethics Committee. The subjects were all recreationally active and were familiar with the experimental procedures used in the study. On test days, subjects were instructed to report to the laboratory in a rested state, having completed no strenuous exercise within the previous 24 h and having abstained from food, alcohol, and caffeine for the preceding 3 h. Testing was conducted at the same time of day ($\pm$ 2 h) for each subject and laboratory visits were separated by at least 48 h.

Experimental overview. All testing was completed at sea level in an air-conditioned laboratory at a temperature of 21°C. The subjects visited the laboratory on six occasions for a 3-wk period to perform exercise tests on an electronically braked cycle ergometer (Lode Excalibur Sport; Lode BV, Groningen, the Netherlands) and a Computrainer cycle ergometry system (Racermate Computrainer, Seattle, WA). Both ergometers were calibrated using a Lode Calibrator 2000 (Lode BV). On the first visit, subjects completed a ramp incremental exercise test to determine $\dot{V}O_{2\max}$ and gas exchange threshold (GET). During the second visit, subjects completed a familiarization 3-min AOT, which was not used in data analysis. This test was repeated during the third laboratory visit to determine EP and WEP. On the fourth visit, subjects performed a 3-min self-paced familiarization trial, which was not used in data analysis. After these preliminary tests, subjects completed two cycling tests. One test comprised enforced-pace CWR at a power output predicted to result in exhaustion in 3 min according to EP and WEP measured in the 3-min AOT (see equation 3 below). The other test was a 3-min self-paced trial. These tests were presented to subjects on separate days in a counterbalanced order.

Exercise tests. The ramp incremental exercise test (INC) consisted of 3 min of pedaling at 0 W, followed by a continuous ramped increase in work rate of 30 W·min⁻¹ until the subject was unable to continue. The subjects were instructed to maintain their preferred cadence (80 rpm, $n = 5$; 90 rpm, $n = 3$), and tests were terminated upon volitional exhaustion or when the required pedal rate could no longer be maintained (i.e., fell by >10 rpm). T_e and peak power output (P_{peak}) were recorded, and $\dot{V}O_{2\max}$ was defined as the highest 30-s mean value achieved. The GET was determined as the first disproportionate increase in carbon dioxide output ($\dot{V}CO_2$) from visual inspection of individual plots of $\dot{V}CO_2$ versus $\dot{V}O_2$ (41). The power output halfway between P_{GET} and P_{peak} (i.e., 50% Δ) was calculated, taking into account the lag time for $\dot{V}O_2$ relative to power output during incremental exercise.

The 3-min AOT comprised 3 min of “unloaded” (i.e., 0 W) baseline pedaling at the subject's preferred cadence followed by the 3-min all-out effort. The resistance on the

pedals during all-out cycling was set using the linear mode of the Lode Excalibur Sport ergometer such that the subject would attain a power output corresponding to 50% Δ on reaching their preferred cadence (linear factor = power/preferred cadence²). Strong verbal encouragement was provided throughout the test, but no information was given regarding elapsed time to prevent pacing. Subjects were instructed to accelerate their cadence to approximately 110–120 rpm over the last 5 s of the baseline period and to attain their peak power output as quickly as possible after the start of the test. They were told to maintain their cadence as high as possible at all times from that point onwards.

EP during AOT was defined as the mean power output over the final 30 s of the test. WEP was calculated as the power–time integral above EP. The power output that would be predicted to lead to exhaustion in 3 min (P_3 ; i.e., the work rate that subjects would maintain during CWR) was then calculated from the algebraically equivalent version of equation 1:

$$P = (W'/T_e) + CP \quad [3]$$

where T_e is the predicted time to exhaustion (i.e., 180 s) and EP and WEP are substituted for CP and W' , respectively. We also used EP and WEP from the 3-min AOT in place of CP and W' (38) in equation 2 to predict T_e for the ramp test that the subjects had already completed ($S = 0.5 \text{ W}\cdot\text{s}^{-1}$). This allowed for a comparison of actual versus predicted T_e to assess the degree to which parameters derived from the 3-min AOT can be used to predict T_e during INC. After returning to the laboratory on the fourth visit for familiarization to self-paced cycling on the Computrainer, subjects completed CWR and SPT in counterbalanced order (four subjects did CWR first and the other four did SPT first) on visits 5 and 6.

CWR (performed on the Lode Excalibur Sport) began with 3 min of 0-W baseline cycling after which work rate was abruptly increased to P_3 . Subjects were instructed to continue for as long as possible at the prescribed cadence (see previous paragraphs), and T_e was recorded to the nearest second with exhaustion defined as a fall in cadence >10 rpm. Subjects were not informed of the work rate, elapsed time, or expected time to exhaustion during this test. SPT was performed on the Computrainer, which provides a reliable measure of power output when guidelines regarding temperature and calibration are followed (8). This ergometer was used for SPT because, unlike the Lode where self-pacing is difficult because the linear factor is fixed for the duration of the test, the Computrainer permits the subject to change gears during exercise. Before each test, the Computrainer was calibrated in accordance with manufacturer recommendations. During each test, a computer screen that displayed the Computrainer software program was placed in front of subjects, allowing them to view a computer-projected simulation of themselves and the distance they had cycled. The test began with 3 min of 0-W baseline cycling after which subjects were asked to perform

as much work as possible within 3 min using a self-selected pacing strategy. Strong verbal encouragement was provided during CWR and SPT.

Before performing AOT, CWR, and SPT, the maximum iEMG of subjects' right musculus vastus lateralis (iEMG_{max}) during cycling was determined. The leg was initially shaved and cleaned with alcohol around the belly of the muscle, and graphite snap electrodes (Unilect; Unomedical Ltd., UK) were adhered to the prepared area in a bipolar arrangement (interelectrode distance: 40 mm). To secure electrodes and wires in place and to minimize movement during cycling, an elastic bandage was wrapped around the subject's leg. Pen marks were made around the electrodes to enable the reproduction of the placement in subsequent tests. For the iEMG_{max} assessment, the test began with 3 min of unloaded baseline pedaling at each subject's preferred cadence, followed by a 5-s all-out sprint with the resistance on the pedals set using the linear mode of the Lode Excalibur Sport ergometer (linear factor = $0.035 \text{ W}\cdot(\text{r}\cdot\text{min}^{-1})^{-2}$). Once this assessment was completed, 5 min of rest was allowed before beginning the subsequent bout. iEMG values recorded during AOT, CWR, and SPT were normalized to the iEMG_{max} that preceded the bout.

Measurements. During all cycling tests, pulmonary gas exchange was measured breath by breath with subjects wearing a nose clip and breathing through a low dead space, low-resistance mouthpiece, and impeller turbine assembly (Jaeger Triple V, Jaeger, Hoechberg, Germany). The inspired and expired gas volume and gas concentration signals were continuously sampled at 100 Hz, the latter using paramagnetic ($\dot{V}\text{O}_2$) and infrared ($\dot{V}\text{CO}_2$) analyzers (Jaeger Oxycon Pro; Jaeger) via a capillary line connected to the mouthpiece. These analyzers were calibrated before each test with gases of known concentration, and the turbine volume transducer was calibrated using a 3-L syringe (Hans Rudolph, Kansas City, KS). The volume and concentration signals were time aligned by accounting for the delay in capillary gas transit and analyzer rise time relative to the volume signal. $\dot{V}\text{O}_2$, $\dot{V}\text{CO}_2$, and $\dot{V}_E$ were displayed breath by breath. HR was measured every 5 s during all tests using short-range radiotelemetry (Polar S610; Polar Electro Oy, Kempele, Finland).

The neuromuscular activity of the musculus vastus lateralis of the right leg was measured using bipolar surface electromyography. The electromyographic signal was recorded using an ME3000PB Muscle Tester (Mega Electronics Ltd., Kuopio, Finland) at a sampling frequency of 1000 Hz. The bipolar signal was amplified (amplifier input impedance > 1 M Ω), and data were collected online in raw form and stored on a personal computer using MegaWin software (Mega Electronics). The raw electromyographic data were subsequently exported as an ASCII file and digitally filtered using Labview 8.2 (National Instruments, Newbury, UK). Initially, the signals were filtered with a 20-Hz high-pass, second-order Butterworth filter to remove contamination from movement artifacts. A linear envelope was then produced via rectification and low-pass (500 Hz) filtering to smooth high-frequency

peaks (e.g., signals in the external environment detected by the electrodes).

During all exercise tests, a blood sample from a fingertip was collected into a capillary tube over the final 20 s of baseline cycling and immediately upon termination of exercise. These samples were subsequently analyzed to determine baseline and end-exercise blood [lactate] (YSI 1500; Yellow Springs Instruments, Yellow Springs, OH). Blood lactate accumulation (Δ blood [lactate]) was calculated as the difference between blood [lactate] at end exercise and blood [lactate] at baseline.

Data analysis procedures. To assess the amount of work performed above CP ($W_{>CP}$; expressed in kJ) during CWR, we used the following equation:

$$W_{>CP} = [(P_3 T_e) - (CP T_e)]/1000 \quad [4]$$

where T_e is the measured time to exhaustion. The total work performed in the SPT (kJ) was calculated as

$$W = (CP \times 180 \text{ s}) + W'/1000 \quad [5]$$

The actual $W_{>CP}$ during SPT we calculated as the power-time integral during the 3-min self-paced cycling bout minus the work performed below CP (i.e., $CP \times 180$ s). To assess $W_{>CP}$ during INC, we calculated the power-time integral for all of the data before exhaustion and subtracted the work performed below CP.

For CWR, SPT, and AOT, baseline $\dot{V}O_2$ was defined as the mean $\dot{V}O_2$ measured over the final 90 s of cycling before the onset of exercise, whereas peak $\dot{V}O_2$ was defined as the highest 30-s mean value recorded during the test. The time-to-attain $\dot{V}O_{2max}$ in the AOT was determined for each individual as the time required for the 5-s rolling-averaged $\dot{V}O_2$ to rise to a value that was within 1 SD of the $\dot{V}O_{2max}$ (using the criterion established in the ramp incremental test) (40). Baseline and end-exercise HR were defined as the mean HR measured over the final 15 s of baseline cycling and exercise, respectively.

Mean iEMG was calculated for 1-s intervals during the 5-s sprint cycling assessment, and iEMG_{max} was defined as the highest 1-s value. Mean iEMG was calculated for 10-s intervals throughout both the baseline and the exercise periods of AOT, CWR, and SPT, and these values were normalized to the iEMG_{max} that preceded the bout. Baseline iEMG was defined as the mean iEMG during the 90 s preceding the onset of exercise, and end-exercise iEMG was defined as the mean iEMG during the final 10 s of exercise. We also calculated peak iEMG, which was defined as the highest 10-s value recorded during the test. Finally, we calculated mean iEMG for 30-s time bins for each experimental condition (i.e., mean = 1–30, 31–60, 61–90, 91–120, 121–150, and 151–180 s) to assess the extent to which iEMG changed as exercise proceeded for the three different protocols.

Statistical analysis. A one-way repeated-measures ANOVA was used to determine the differences between trials in relevant physiological and performance variables. Where the analysis revealed a significant difference, individual

paired *t*-tests were used to determine the origin of such effects. Paired *t*-tests were also used to identify significant differences between iEMG at 30-s and iEMG at 180-s for each condition. Paired *t*-tests and intraclass correlation coefficients were used to examine the relationship between actual and predicted T_e for INC and CWR and actual and predicted performance for SPT. The coefficient of variation (CV) was calculated for each individual as the standard deviation of the predicted and actual measure relative to the mean of the predicted and actual measure and expressed as a percentage. All data are presented as mean $\pm$ SD. Statistical significance was accepted when $P < 0.05$.

RESULTS

The $\dot{V}O_{2max}$ for INC was $4.24 \pm 0.69 \text{ L} \cdot \text{min}^{-1}$ with GET occurring at $1.80 \pm 0.31 \text{ L} \cdot \text{min}^{-1}$. P_{peak} and P_{GET} were 376 ± 57 and $117 \pm 38 \text{ W}$, respectively. EP and WEP from the AOT were $260 \pm 60 \text{ W}$ and $16.5 \pm 4.0 \text{ kJ}$, respectively. P_3 (i.e., the work rate for CWR estimated via equation 3) was $345 \pm 58 \text{ W}$. The actual T_e for CWR was $185 \pm 24 \text{ s}$ (range = 145–217 s), which was not significantly different from 180 s (i.e., the predicted value), and the actual and the predicted T_e for CWR were significantly correlated ($r = 0.99$, $P < 0.01$). The CV between predicted and actual T_e in the CWR was $8\% \pm 5\%$. The actual T_e for INC (753 ± 121 ; range = 622–939 s) was not significantly different from the predicted T_e ($754 \pm 122 \text{ s}$; range = 605–943 s), and these values were also highly correlated ($r = 0.92$, $P < 0.01$; see Fig. 1). The CV between predicted and actual T_e in the INC was $3\% \pm 3\%$. The actual total work performed in SPT ($63.1 \pm 10.6 \text{ kJ}$) was not significantly different from the predicted total work performed ($62.6 \pm 10.4 \text{ kJ}$), and the actual and predicted values were significantly correlated ($r = 0.94$, $P < 0.01$). The CV between predicted and actual work performed in the SPT was $3\% \pm 3\%$.

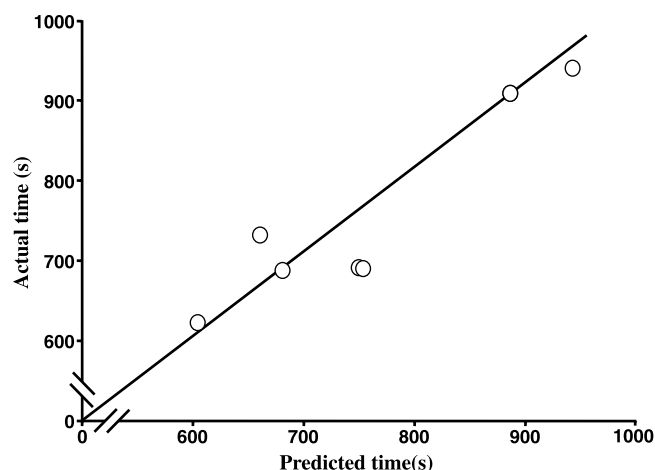


FIGURE 1—Actual versus predicted T_e for INC ($r = 0.92$ for $n = 8$, $P < 0.01$). Prediction was made using parameters derived from a 3-min AOT (i.e., EP and WEP).

$W_{>CP}$ values for CWR and SPT (16.6 ± 7.4 and 15.3 ± 5.6 kJ, respectively) were not different from one another or from WEP for AOT (16.5 ± 4.0 kJ, $P > 0.05$). Furthermore, these values were not different ($P > 0.05$) from $W_{>CP}$ for INC (16.4 ± 4.0 kJ). The CV for $W_{>CP}$ among conditions was $19\% \pm 19\%$ between AOT and INC, $17\% \pm 25\%$ between AOT and CWR, and $12\% \pm 11\%$ between AOT and SPT. This similarity of the capacity to perform work above CP for four different protocols is depicted for a representative subject in Figure 2. A schematic representation of the different ways in which this capacity was depleted along with the associated power profiles is provided in Figure 3.

Table 1 presents physiological responses to AOT, CWR, and SPT. The peak $\dot{V}O_2$ values for these three protocols were not different from one another or from $\dot{V}O_{2max}$ for INC ($P > 0.05$). The CV for $\dot{V}O_{2max}$ among conditions was $5\% \pm 4\%$ between INC and AOT, $4\% \pm 2\%$ between INC and CWR, and $3\% \pm 1\%$ between INC and SPT. However, the time to achieve the peak $\dot{V}O_2$ was shorter for AOT compared with the other two conditions, which did not differ (Table 1). Group mean $\dot{V}O_2$ response profiles for AOT, CWR, and SPT are shown in Figure 4. There was no significant difference in end-exercise blood lactate concentration or end-exercise iEMG between conditions; however, iEMG profiles during the bouts were different. Specifically, peak iEMG was greater for AOT and iEMG decreased by 44% from the initial to the final 30 s of the test. In contrast, iEMG increased from the initial to the final 30 s of CWR and

SPT (by 19% and 15%, respectively; Table 1). Group mean iEMG response profiles are shown in Figure 5.

DISCUSSION

The results of this investigation indicate that the total amount of work that can be performed before the limit of tolerance during exercise exceeding the CP (i.e., $W_{>CP}$) is similar across different high-intensity exercise protocols that expend the W' at different rates. Specifically, in agreement with our first hypothesis, regardless of whether it was produced by enforced-pace exercise (either INC or CWR) or a protocol where subjects were allowed to choose their own pacing strategy (SPT), the termination of exercise coincided with a similar capacity for $W_{>CP}$. This is consistent with the CP model of human bioenergetics, which proposes that the power–duration relationship for high-intensity exercise is defined by two constants: a power asymptote (CP) and a curvature constant (W'), with the latter (expressed in joules) representing the finite capacity for $W_{>CP}$. The $W_{>CP}$ across SPT, CWR, and INC was similar to the total amount of work that subjects could complete above EP during the 3-min AOT. Together, the CP and W' estimated using the AOT enabled the prediction of the T_e during CWR with moderate accuracy (CV $\sim 8\%$) and the prediction of T_e during INC and the total work performed during SPT with good accuracy (CV $\sim 3\%$ for both).

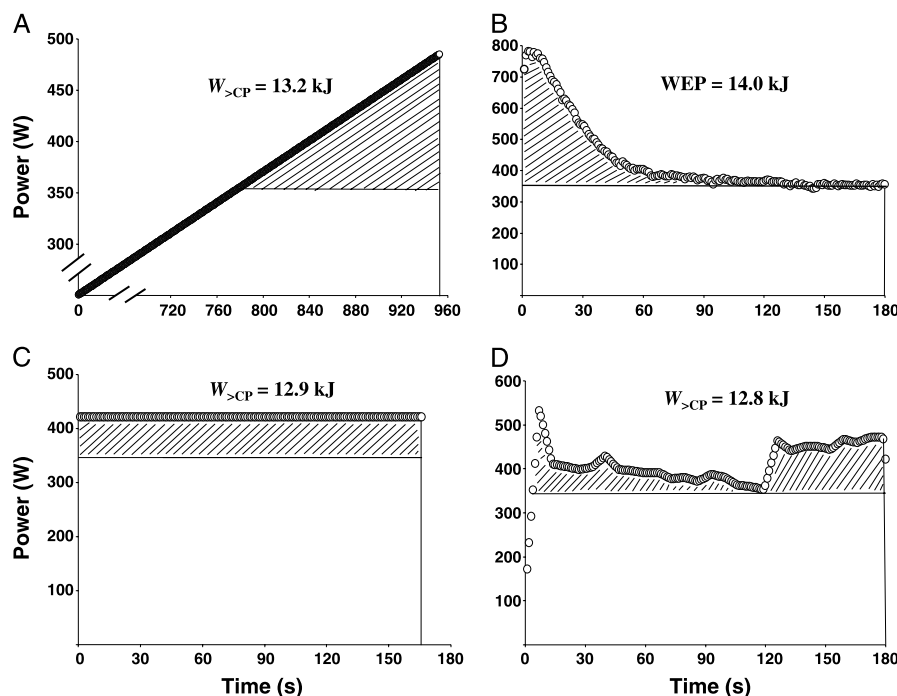


FIGURE 2—Power profile for a representative subject during INC (A), AOT (B), CWR (C), and SPT (D). The horizontal line superimposed on each panel indicates the subject's CP (i.e., as estimated by EP from AOT; e.g., 353 W for this subject), and the striped area represents the work the subject performed above that power output for each protocol. Notice the similarity of this capacity for $W_{>CP}$ for four distinctly different protocols.

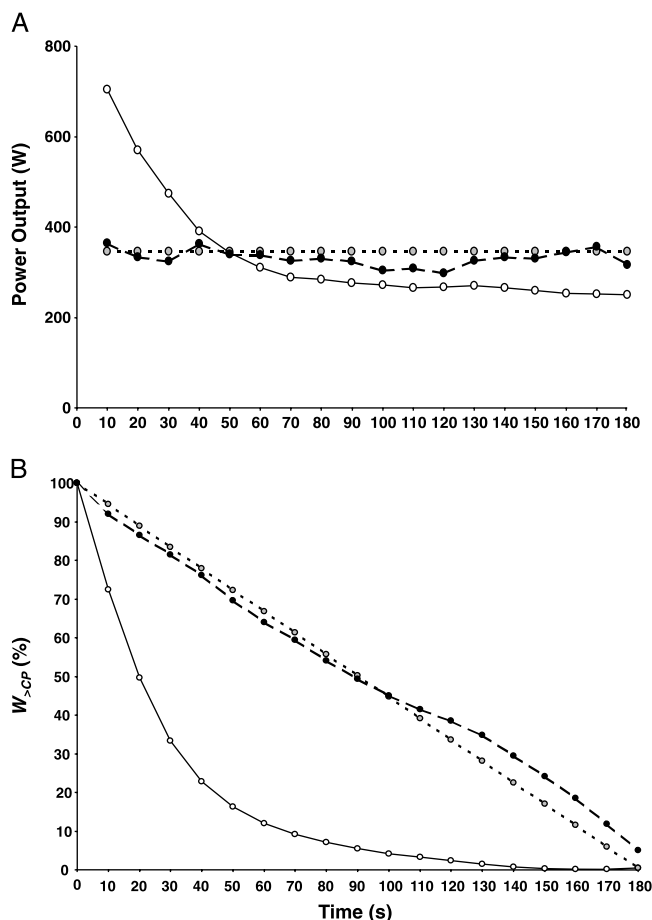


FIGURE 3—A. The power profiles for AOT (open circles), CWR (closed circles with gray fill), and SPT (closed circles with black fill). **B.** A schematic representation of depletion of the capacity for $W_{>CP}$ for the same three conditions. These depictions reflect group mean data for work performed at 10-s intervals throughout the exercise bouts. Error bars have been excluded for clarity. Notice how the end of exercise coincides with complete depletion/use of a finite capacity for work regardless of the power profile. See text for further details.

The precise determinants of W' (i.e., the factors that dictate the capacity for $W_{>CP}$) are not yet well understood. According to the classical interpretation, it represents an “anaerobic work capacity”; that is, it is linked to the ability to derive energy from high-energy phosphate hydrolysis and anaerobic glycolysis with a small contribution from myoglobin- and

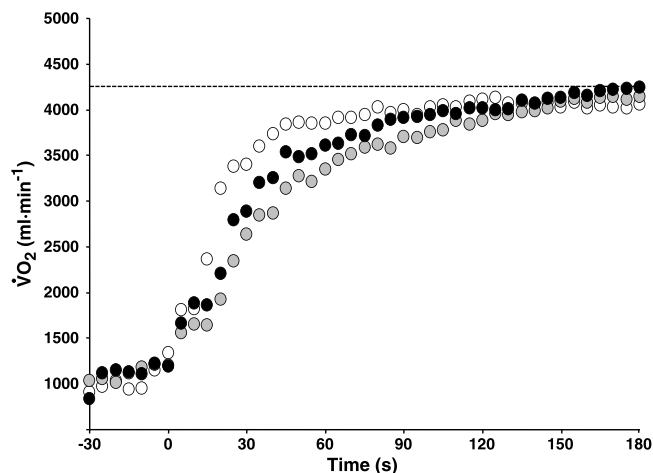


FIGURE 4—Group mean $\dot{V}O_2$ response profiles for AOT (open circles), CWR (closed circles with gray fill), and SPT (closed circles with black fill). Dashed horizontal line indicates $\dot{V}O_{2max}$ from INC. Notice that despite the different protocols that were used, $\dot{V}O_2$ reached a peak value that was not different from $\dot{V}O_{2max}$.

hemoglobin-bound $\dot{V}O_2$ stores (10,25–27). However, this is now understood to be an oversimplification (16). An alternative view is that W' is related to the attainment of some critical level of metabolite accumulation (e.g., $[H^+]$ and/or $[P_i]$ and/or $[ADP]$) (7,12,17,18,39). Interestingly, CP represents the highest external power output that can be maintained without a continued rise in $\dot{V}O_2$ to the $\dot{V}O_{2max}$ (16,33). Therefore, another feature of CWR and INC performed above CP is that a reproducible $\dot{V}O_{2max}$ should be attained when the W' has been expended.

In the present study, we used the parameters of the power–duration relationship for high-intensity exercise to predict the CWR that subjects could sustain for 3 min (see equation 3) and the total work that could be performed in the SPT. We also predicted in retrospect the amount of time subjects could sustain an INC test at a given ramp slope (i.e., $0.5 \text{ W} \cdot \text{s}^{-1}$; see equation 2). However, instead of deriving these parameters in the conventional manner (i.e., constructing the power–duration hyperbola by plotting T_c from multiple CWR bouts performed to exhaustion at different work rates), we estimated them from the data obtained during a single 3-min AOT. We have noted previously that in some individuals, the ability of the AOT parameters to predict T_c during

TABLE 1. Physiological parameters for 3-min AOT, CWR test with predicted duration of 3 min, and 3-min self-paced time trial (SPT).

	AOT	CWR	SPT
Baseline HR (beats·min ⁻¹)	95 ± 10	98 ± 7	92 ± 20
End-exercise HR (beats·min ⁻¹)	174 ± 4	177 ± 8	171 ± 7
Baseline $\dot{V}O_2$ (L·min ⁻¹)	0.95 ± 0.25	1.06 ± 0.21	1.10 ± 0.81
Peak $\dot{V}O_2$ (L·min ⁻¹)	4.10 ± 0.79	4.14 ± 0.75	4.20 ± 0.77
Time to peak $\dot{V}O_2$ (s)	59 ± 24*	140 ± 22	136 ± 24
Baseline blood lactate concentration (mM)	1.6 ± 0.4	2.1 ± 0.4	1.8 ± 0.8
End-exercise blood lactate concentration (mM)	8.7 ± 1.7	7.6 ± 1.8	8.2 ± 1.7
Δ Blood lactate concentration (mM)	7.1 ± 1.4	5.4 ± 1.9	6.3 ± 2.2
Baseline iEMG (% iEMG _{max})	8 ± 3	10 ± 5	7 ± 7
End-exercise iEMG (% iEMG _{max})	39 ± 6	48 ± 12	47 ± 6
Peak iEMG (% iEMG _{max})	74 ± 11**	51 ± 15	50 ± 6

*Significantly different from CWR and SPT ($P < 0.05$).

**†Significantly different from SPT ($P < 0.05$).

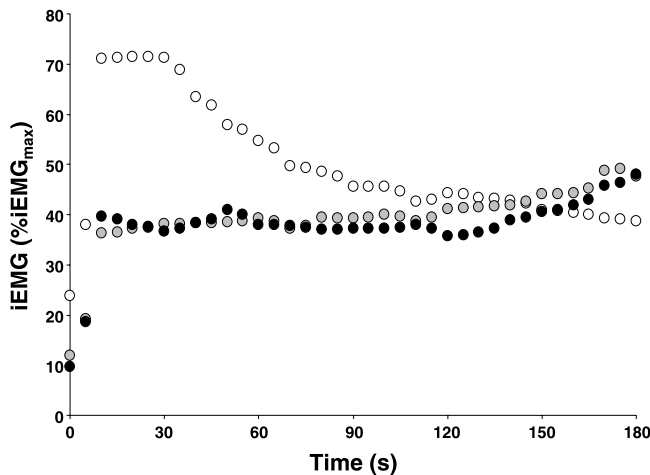


FIGURE 5—Mean iEMG response expressed relative to $iEMG_{max}$ for AOT (open circles), CWR (closed circles with gray fill), and SPT (closed circles with black fill). Notice how iEMG reaches a higher peak but subsequently decreases during all-out exercise compared with the increase in iEMG with time that is observed for the other two protocols. Notice also that the iEMG profile is similar in the CWR (enforced-pace) and SPT (self-paced) tests. Values represent 10-s mean values expressed as a percentage of the highest 1-s iEMG measured during a preceding 5-s sprint.

CWR exercise may be less accurate at higher compared with lower severe-intensity work rates (39). This may be partly due to relatively poor test–retest reliability of W' compared with CP whether established using conventional or all-out protocols (3,14,15,36) as well as the greater relative contribution of W' to total work performed in shorter exercise bouts. In the present study, the accuracy of the predicted T_e in CWR, which was performed at approximately 90% Δ , was moderately good ($CV = \sim 8\%$), whereas the predictions of T_e and performance in INC and SPT were quite precise (CV of $\sim 3\%$). This largely confirms our second hypothesis and provides further support for the notion that the EP and WEP during 3 min of all-out cycling can be used to estimate CP and W' , respectively (3,37,38).

Our ability to use estimates of CP and W' to predict T_e during CWR and INC and performance during SPT is consistent with previous research, which shows that $W_{>CP}$ is similar regardless of the chosen work rate(s) above CP or the forcing function by which the work rate(s) is applied. For example, it has been established that $W_{>CP}$ is similar for CWR exercise, INC exercise, and exercise where step-wise variations of power output above CP are present (6,12,14,28,29). However, it is important to note that pacing was enforced during these previous investigations; that is, the investigator controlled the external power output(s) that the subject was expected to produce throughout the test. This is unlike athletic competition where the pacing strategy is selected by the subject to achieve the optimal performance (35). This point is relevant when investigating $W_{>CP}$ because it has been suggested that alterations in power output that occur during self-paced exercise are established through central neural control. Specifically, it has been suggested that the brain regulates muscle activation in response to

feedback information from peripheral receptors and feed-forward anticipation of ongoing events (20,24,31,32). This “central governor” would presumably be related in some way to the cascade of events that characterize the limit of tolerance (i.e., $\dot{V}O_2$ failing to continue to rise to a higher value and the inability to complete further $W_{>CP}$). If true, it is possible that affording subjects greater autonomy over pacing strategy during high-intensity exercise might reduce the sensations of fatigue and allow performance to be enhanced (23). Interestingly, self-pacing during 5000-m rowing time trials has been shown to result in similar performance as an enforced-pace protocol while producing less homeostatic disturbance (e.g., less blood lactate accumulation and temperature elevation) (21).

In the present study, we asked the subjects to cycle as far as possible in 3 min on an ergometer that provided feedback on distance covered. This allowed us to compare $W_{>CP}$ during a self-paced exhaustive 3-min bout of high-intensity cycling (SPT) with $W_{>CP}$ during an enforced-pace approximately 3-min (i.e., 185 ± 24 s) CWR test. Consistent with our first hypothesis, $W_{>CP}$ during SPT was not significantly different from $W_{>CP}$ during CWR. Furthermore, this value was also similar to $W_{>CP}$ during INC and the WEP measured during the AOT. Our study design also allowed us to compare the consistency of the peak $\dot{V}O_2$ across the different exercise protocols. The peak $\dot{V}O_2$ during SPT was not significantly different from the peak $\dot{V}O_2$ during CWR, and this value was also similar to the $\dot{V}O_{2max}$ from INC and the peak $\dot{V}O_2$ measured during AOT. These latter results are consistent with our previous study (5) in which we reported that there was no difference in the peak $\dot{V}O_2$ attained during INC when subjects were allowed to self-pace compared with when the work rate increment was linear and controlled externally. The findings of the present study indicate that the functional link between the depletion of a reproducible capacity for $W_{>CP}$, the attainment of a reproducible $\dot{V}O_2$ peak, and, ultimately, the factors that lead to the termination of exercise during high-intensity exercise is not affected when subjects are allowed to self-pace.

Although the cause(s) of fatigue during supra-CP exercise may ultimately be the same, the pattern of pacing selected in SPT (i.e., greater than mean power output at the beginning and end of exercise) resulted in a different profile of $W_{>CP}$ expenditure compared with CWR, as illustrated in Figure 3. In CWR, the $W_{>CP}$ expenditure was linear throughout exercise, whereas in SPT, it fell more steeply over the first approximately 100 s and then less steeply for most of the remaining approximately 80 s. It is possible that this self-selected pacing strategy allows subjects to feel more comfortable over a greater period of the exercise trial (i.e., more of the total available $W_{>CP}$ are remaining from ~ 100 to ~ 170 s), while still permitting a similar $W_{>CP}$ to be accomplished. This may explain the lower RPE but similar performance reported by Lander et al. (21) during a self-paced compared with an enforced-pace 5000 m rowing time trial. A relatively higher power output at the beginning of

high-intensity exercise (i.e., a fast-start pacing strategy) also speeds $\dot{V}O_2$ kinetics (see Fig. 4) and may benefit performance (1,2,19). Interestingly, although the $W_{>CP}$ was not significantly different between any of the protocols investigated in the present study, it was lowest for SPT (15.3 kJ compared with ~ 16.5 kJ for the other tests). It is possible that this was a function of the subjects' relative lack of familiarity with the SPT protocol because they completed only one familiarization trial. It is known that pacing strategy can be optimized with additional experience of the exercise task (22,24).

We used iEMG to assess muscle activation during the different protocols in the present study. There are limitations to inferring muscle activation from iEMG during cycling (especially when cadence is not constant; e.g., during AOT). Moreover, the iEMG results in Figure 5 represent the mean of a 10-s sample as a proportion of the highest 1-s value recorded during an initial 5-s sprint cycling bout. Therefore, it is only possible to make qualitative comparisons between the iEMG profiles recorded in the different protocols. End-exercise iEMG was not significantly different for AOT, CWR, and SPT. However, iEMG reached an early peak during AOT, after which it declined, whereas CWR and SPT were characterized by a progressive increase in iEMG until the termination of exercise. The muscle activation profile in AOT would allow $W_{>CP}$ to be expended rapidly (see Fig. 3B), with power output falling precipitously towards its nadir as the $W_{>CP}$ becomes exhausted (see Fig. 3A). Conversely, during CWR and SPT, $W_{>CP}$ is expended at a slower rate (Fig. 3B), and muscle activation increases (Fig. 5) to maintain the same external power output in the face of mounting fatigue. Importantly, the similarity of the iEMG profile for CWR and SPT (Fig. 5 and Table 1) suggests that affording subjects the autonomy to pace their exercise efforts

does little to change the sequence of events that defines performance capacity matched-duration during high-intensity exercise (see Figs. 2D, 3, and 4).

In conclusion, the results of this study indicate that the use of a "fixed" capacity for $W_{>CP}$ (i.e., W') and the associated increase in $\dot{V}O_2$ to a reproducible peak $\dot{V}O_2$ (i.e., $\dot{V}O_{2max}$) are consistently associated with the attainment of the limit of tolerance during high-intensity ($>CP$) exercise. Specifically, we have shown that high-intensity cycling that involves different profiles of external power production will cause $W_{>CP}$, iEMG, and $\dot{V}O_2$ to change at different rates; ultimately, however, performance in all these conditions is associated with a "complete" depletion of W' , as indicated by similar $W_{>CP}$ across all exercise trials. Although affording subjects the ability to expend this capacity of their own accord by selecting their preferred pacing strategy may alter the profiles of $W_{>CP}$ expenditure and $\dot{V}O_2$ kinetics, and perhaps perception of effort, it does not alter either the capacity for $W_{>CP}$ or the peak $\dot{V}O_2$ that can be attained. Finally, the CP and W' parameters estimated in the 3-min AOT enabled the prediction of time to exhaustion and performance in INC and SPT with high accuracy (CV $\sim 3\%$) and the prediction of time to exhaustion during CWR test with moderate accuracy (CV $\sim 8\%$). This test also elicits a peak $\dot{V}O_2$ response that is not different from $\dot{V}O_{2max}$. Consequently, this single test can be used to determine the three physiological parameters (i.e., CP, W' , and $\dot{V}O_{2max}$) that together determine high-intensity exercise tolerance.

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